MANCHESTER 1824
The University of Manchester

AERO60492 – Autonomous Mobile Robots

Lecture 2 – Encoded Data Communications

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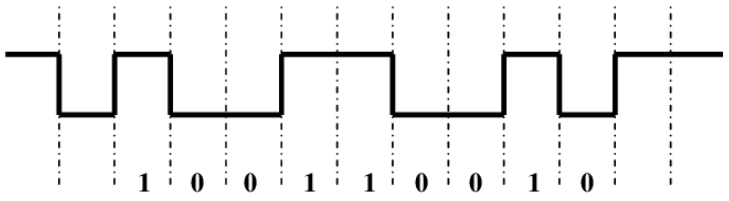
Aims

- Understand what serial data communications are
- What types are commonly found in robotics
- Bits and bytes
- Checksums
- Interpret binary data frame specification
  - implement binary decoder



Coursework 1

What is data

- Transfer of information
- For mobile robots this is measurements and commands
  - Usually binary format
  - Sometimes plain text

e.g. NMEA

Inputs from sensors

e.g. PWM

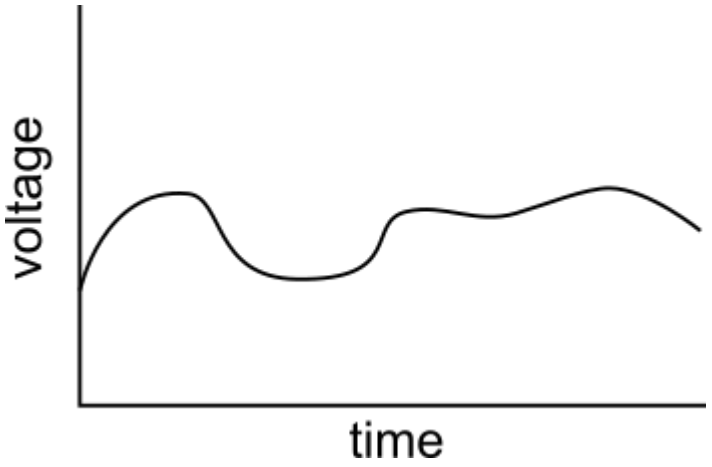
Outputs to actuators

Example of Ardupilot running on Pixhawk style hardware

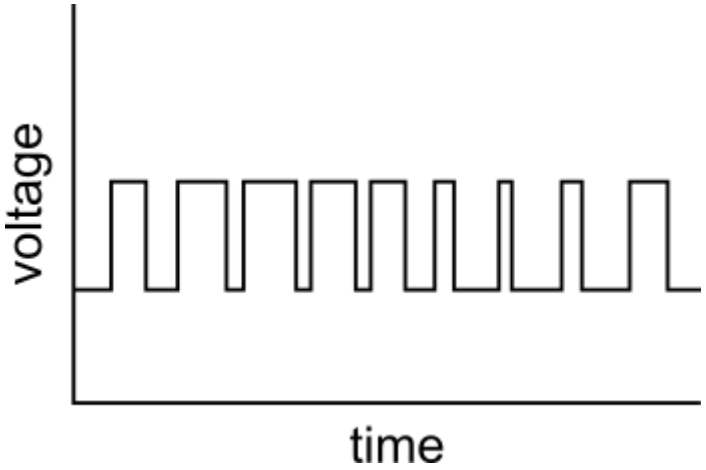
What is serial data

- Microcontrollers/processors fundamentally measure and create voltages as their input and output.

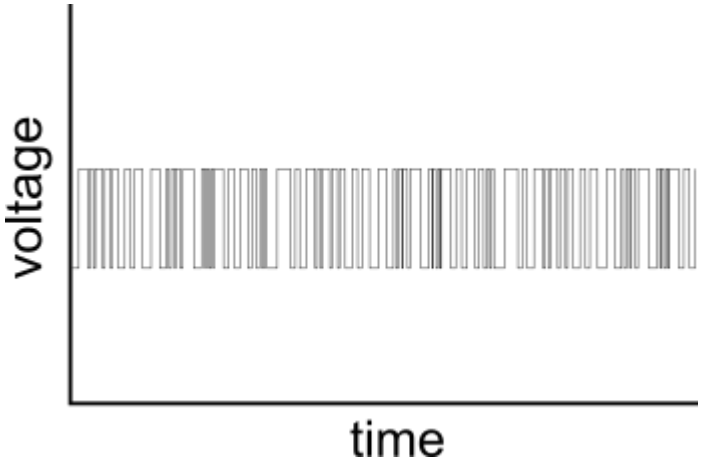
Analogue
Value mapped to voltage



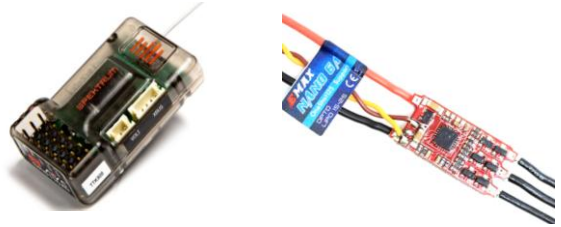
PWM
Value mapped to timings



Digital
Value(s) encoded in binary



Types of serial data on mobile robotics



Proprietary/custom

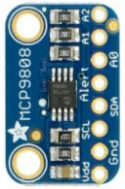
UART

USB

SPI

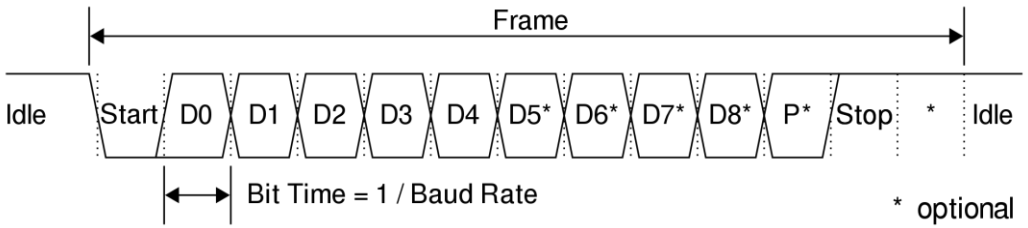
CANBUS

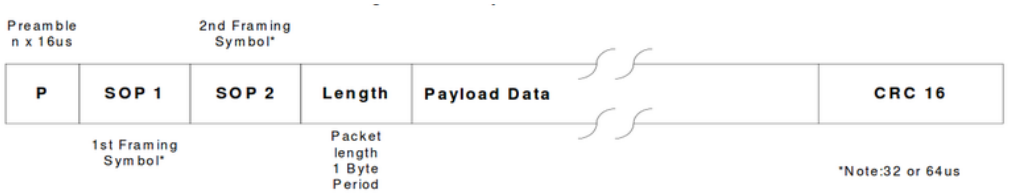
I2C

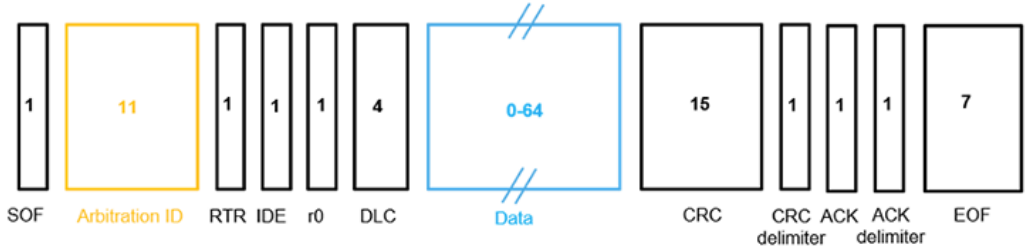


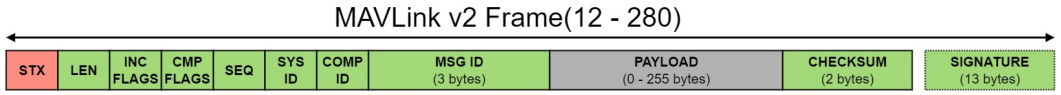
They all need a driver

- Converts pattern of voltages into numeric data
- Two stages
  - Hardware layer
    voltage ⟷ binary
    Defined by serial data protocol
  - Software layer
    binary ⟷ numerical
    Defined by serial frame/byte protocol









Bits and Bytes

- All computing based on splitting data into groups of 8 bits
- 8-bits = 1 byte (4-bits = nibble)
- 1 byte provides 256 unique combinations of its bits
  - $2^{(\text{bit-depth})}$
- There are several human readable representations
- Binary -> decimal interpretation

Binary	[0	0	0	1	1	0	0	1]
	d_7	d_6	d_5	d_4	d_3	d_2	d_1	d_0

$$D = d_0 \times 2^0 + d_1 \times 2^1 \; d_2 \times 2^2 \; d_3 \times 2^3 \; d_4 \times 2^4 \; d_5 \times 2^5 \; d_6 \times 2^6 \; d_7 \times 2^7$$

D = 1 + 8 + 16 = 25

Decimal representation of byte

- Get a calculator:

$$D = d_0 \times 2^0 + d_1 \times 2^1 + d_2 \times 2^2 + d_3 \times 2^3 + d_4 \times 2^4 + d_5 \times 2^5 + d_6 \times 2^6 + d_7 \times 2^7$$

[0 0 1 1 0 0 0 1] = 49

[1 0 1 1 0 0 1 1] = 179

[0 1 0 0 0 0 0 1] = 65

Text representation of byte

- Assign unique letters to each value.
- 256 options covers entire English alphabet and numbers, plus a range of characters
- Decimal 54 = ‘6’
- Decimal 121 = ‘y’

Dec	Hx	Oct	Char		Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL	(null)	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH	(start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX	(start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX	(end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT	(end of transmission)	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ	(enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK	(acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL	(bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS	(backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB	(horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF	(NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT	(vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF	(NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR	(carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO	(shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI	(shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE	(data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1	(device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2	(device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3	(device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4	(device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK	(negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN	(synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB	(end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN	(cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM	(end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB	(substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC	(escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS	(file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	GS	(group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS	(record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US	(unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	DEL

https://www.asciitable.com/

Text representation of byte

43 =

65 =

10 =

Some characters are not human readable. They are instructions to a text editor.

72 101 108 108 111 = ?

Dec	Hx	Oct	Char		Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL	(null)	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH	(start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX	(start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX	(end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT	(end of transmission)	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ	(enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK	(acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL	(bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS	(backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB	(horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF	(NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT	(vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF	(NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR	(carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO	(shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI	(shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE	(data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1	(device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2	(device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3	(device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4	(device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK	(negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN	(synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB	(end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN	(cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM	(end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB	(substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC	(escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS	(file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	GS	(group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS	(record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US	(unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	DEL

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Hexadecimal representation

- Human friendly representation of all values of a byte
- Uses less written characters than decimal/ASCII
- Easier than binary to read
- An efficient half-way representation
- Independent of language
- Assign a number/letter to each 4 bits (a nibble)
  - 4 bits have 16 combinations
  - 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

Hexadecimal representation

Decimal 0-15

Hex 0-F

Decimal 0-255

Hex 00-FF

Might be written 0xD3

Ignore this bit. It is just shorthand to indicate the hexadecimal format has been used

This is the value

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	DEL

Source: www.LookupTables.com

Larger values

- What about values outside than the range 0-255?
- 16-bit number = 2 bytes

$$[\underbrace{0\ 0\ 1\ 1\ 0\ 0\ 0\ 1}_{\text{byte 1}}\ \underbrace{1\ 0\ 0\ 0\ 1\ 1\ 1\ 0}_{\text{byte 0}}] = 12686$$

Even larger values

- Even bigger numbers
- 64-bit number = 8 bytes

binary [1 0 1 1 1 1 1 1 0 0 1 1 0 0 0 0 1 1 0 0 0 1 1 0 0 0 1 1 1 0 1 0 0 0 0 0 1 1 1 1 1 1 0 0 1 1 0 0 0 0 1 1 0 0 0 1 1 0 0 0 1 1 1 0] = 13,776,729,212,801,790,350

	byte 7	byte 6	byte 5	byte 4	byte 3	byte 2	byte 1	byte 0
decimal	95	48	198	58	15	204	49	142
hex	5F	30	C6	3A	0F	CC	31	8E
ASCII lookup	‘b’	‘0’	‘Æ’	‘:’	<SI>	‘Ì’	‘1’	‘Ž’

In the quintillion ranges

‘Shift-In’ command. Not representable by a displayed character

Significant values

- Byte 0 known as the least significant byte
- Byte N known as the most significant byte

binary [1 0 1 1 1 1 1 1 0 0 1 1 0 0 0 0 1 1 0 0 0 1 1 0 0 0 1 1 1 0 1 0 0 0 0 0 1 1 1 1 1 1 0 0 1 1 0 0 0 0 1 1 0 0 0 1 1 0 0 0 1 1 1 0] = 13,776,729,212,801,790,350

	byte 7	byte 6	byte 5	byte 4	byte 3	byte 2	byte 1	byte 0
decimal	95	48	198	58	15	204	49	142

MSB
Most significant byte

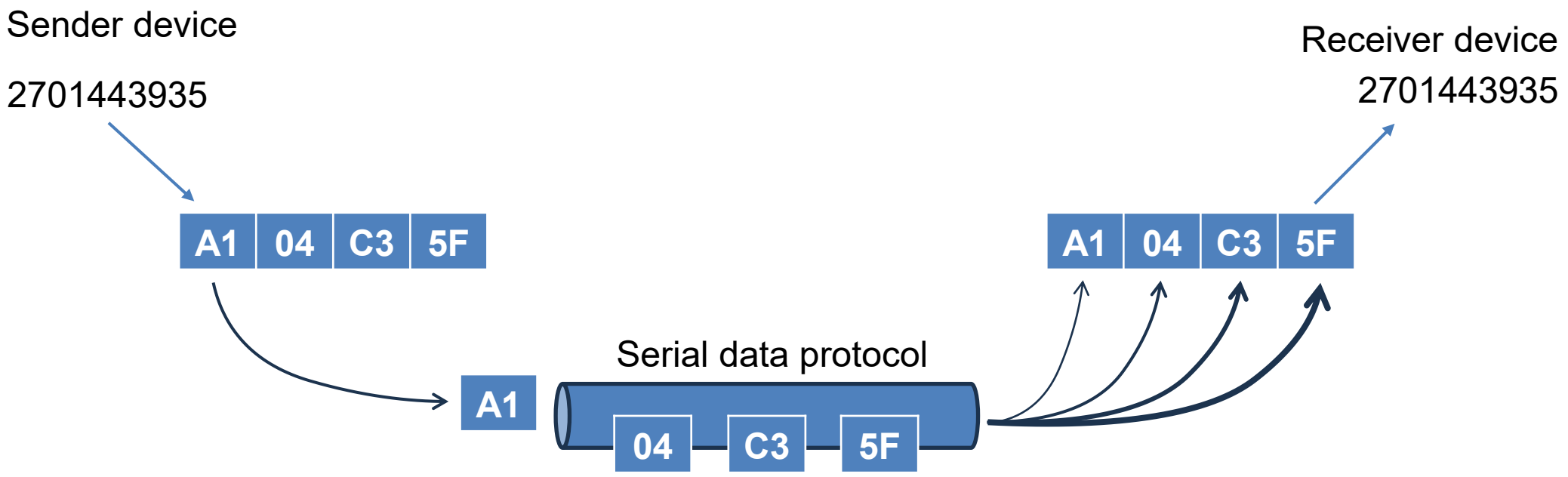
LSB
Least significant byte

Endianness

- We can break down large numbers into byte sized components
- We can then send them one at a time over some communication method (UART, SPI, CAN, etc)
- Do we start with byte 0 or byte N?
- The order is known as endianness and is essential that both ends of the communication chain use the same order
- Big endian ordering because the byte which has the most significant impact on the value is first. Byte N
- Little endian is the opposite starting with the byte which has the last effect on the overall magnitude of value. Byte 0

Sending serial data

- Most communication systems are serial-based
- They split values into bytes and send them one at a time



Sending serial data

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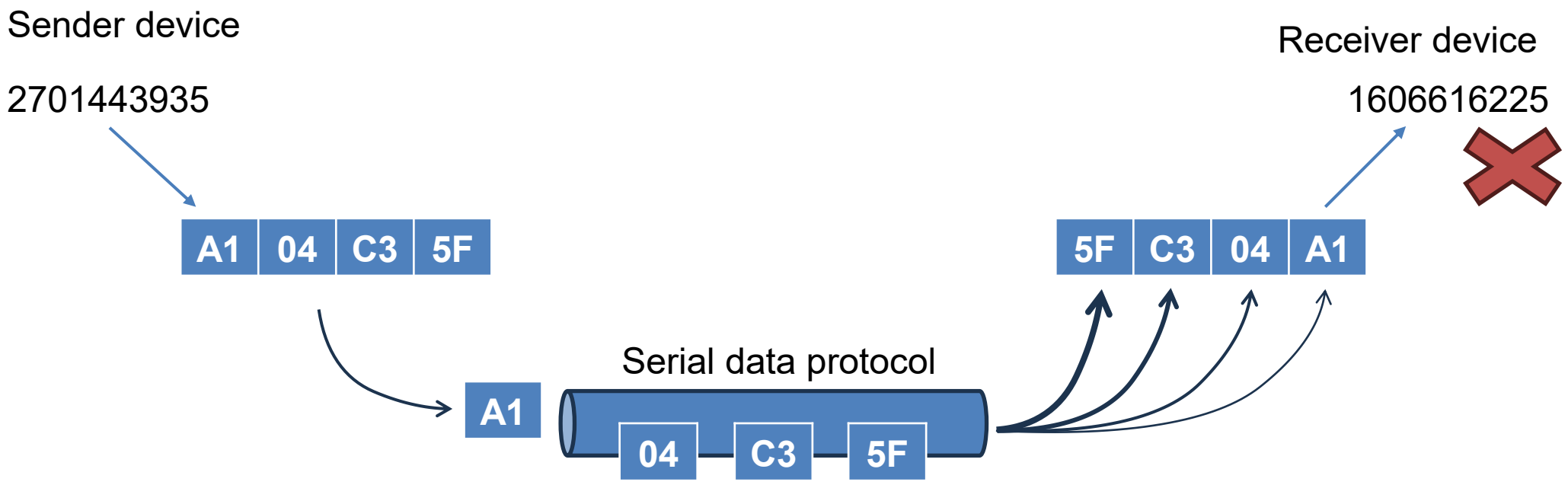

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Serial Data Frames

- Multiple datum packaged together into a frame
- Why:
  - for timings
  - for segregation
  - for efficiency
- Additional stage separating bytes into groups
- Pattern given in a serial data frame definition
- Further stages using functions/lookup tables or unit conversions

Some examples

MAVLink v2 Frame(12 - 280)

STX	LEN	INC FLAGS	CMP FLAGS	SEQ	SYS ID	COMP ID	MSG ID (3 bytes)	PAYLOAD (0 - 255 bytes)	CHECKSUM (2 bytes)	SIGNATURE (13 bytes)

Message frame definition used for the MAVLink protocol

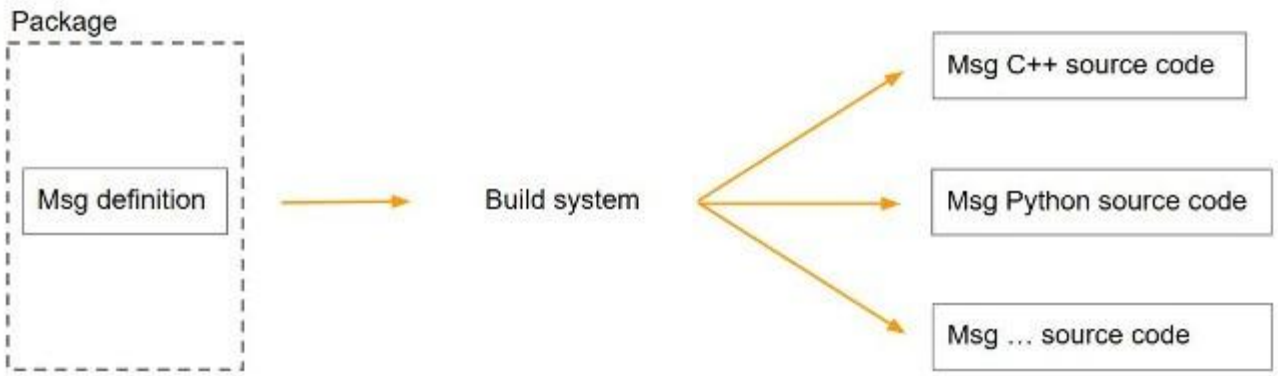
ESC single wire 19200BPS serial data version

Bale No	Bale No	Rx throttle	Rx throttle	Actual output throttle	Actual output throttle	Electric RPM	Electric RPM	Busbar Voltage	Busbar Voltage	Busbar Current	Busbar Current	Phase Wire current	Phase Wire current	MOS Temp	Capacitance Temp	Status Code	Status Code	Verify Code	Verify Code
OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX	OxXX
0-65535,use hex system		0-1024				0-65535		0-65535		±32767				0-255		0-65535,use binary system		0-65535,use hex systems	
XXXX,use decimalism		XXXX*100/1024 floating-type,use one decimal point				XXXX*10/(Tooth *Poles),Input from phone and use integer		XXXX/10,use one decimal point		XXXX transfer to signed integer XXXX/64,use one decimal point				ADC value match with table look-up's value		XX show decima		Sum Verification,2 2 data bytes sum=XXXX	

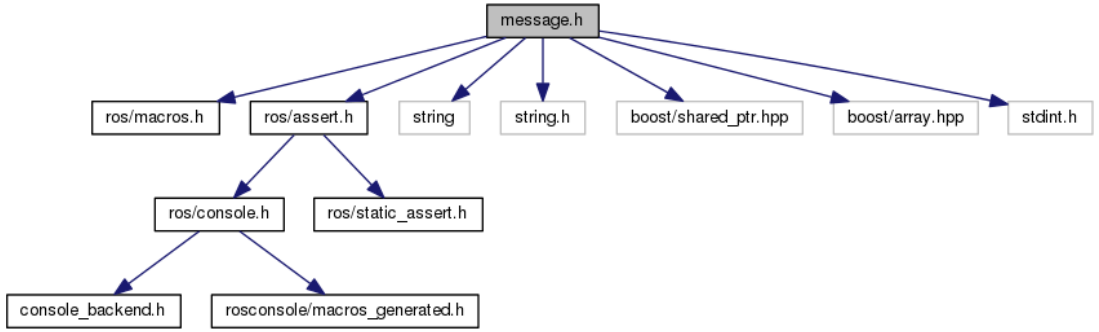
Message frame definition for the T-Motor Alpha ESC

Some examples

https://wiki.ros.org/msg

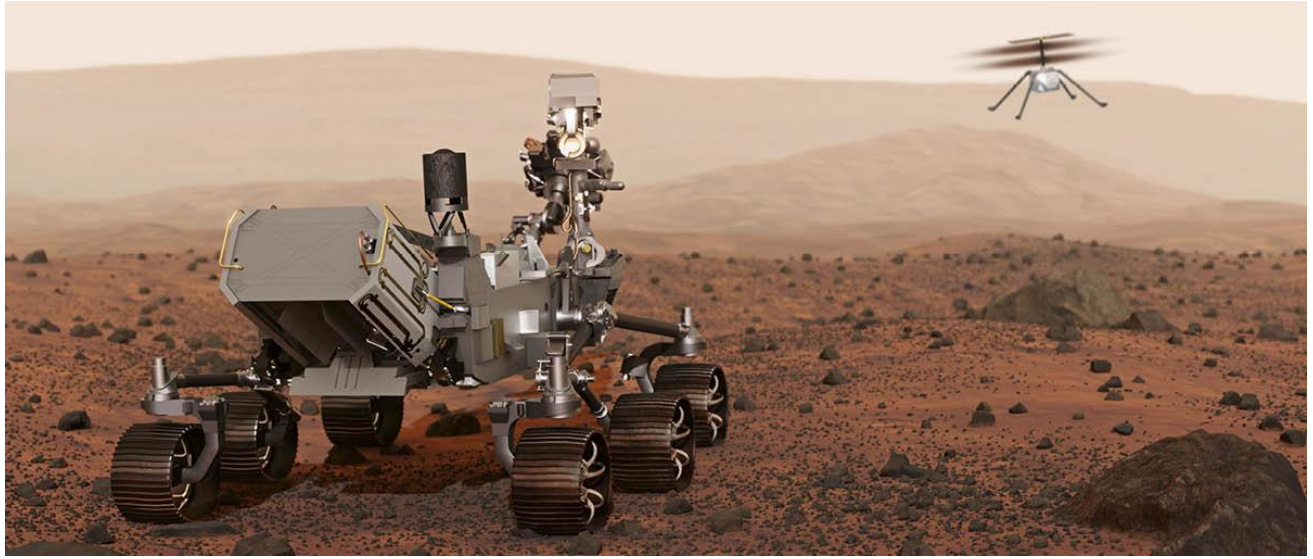


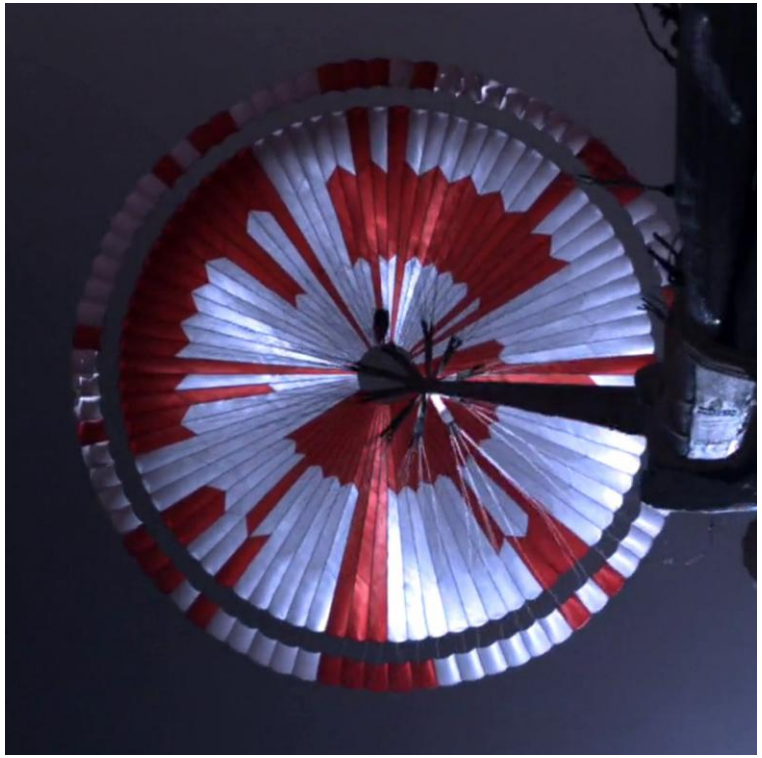
Higher level specification



Auto-generates code libraries

A challenge





binary?

Tutorial

Tutorial 1 – Interpreting Serial Frame Specifications

Answers provided next week in recorded lecture.

Coursework 1

- Set later today
- Coursework specification sheet and rubric released on canvas
- Check file allocation!

- Advice to practice basic Python tutorial
- https://www.w3schools.com/python/